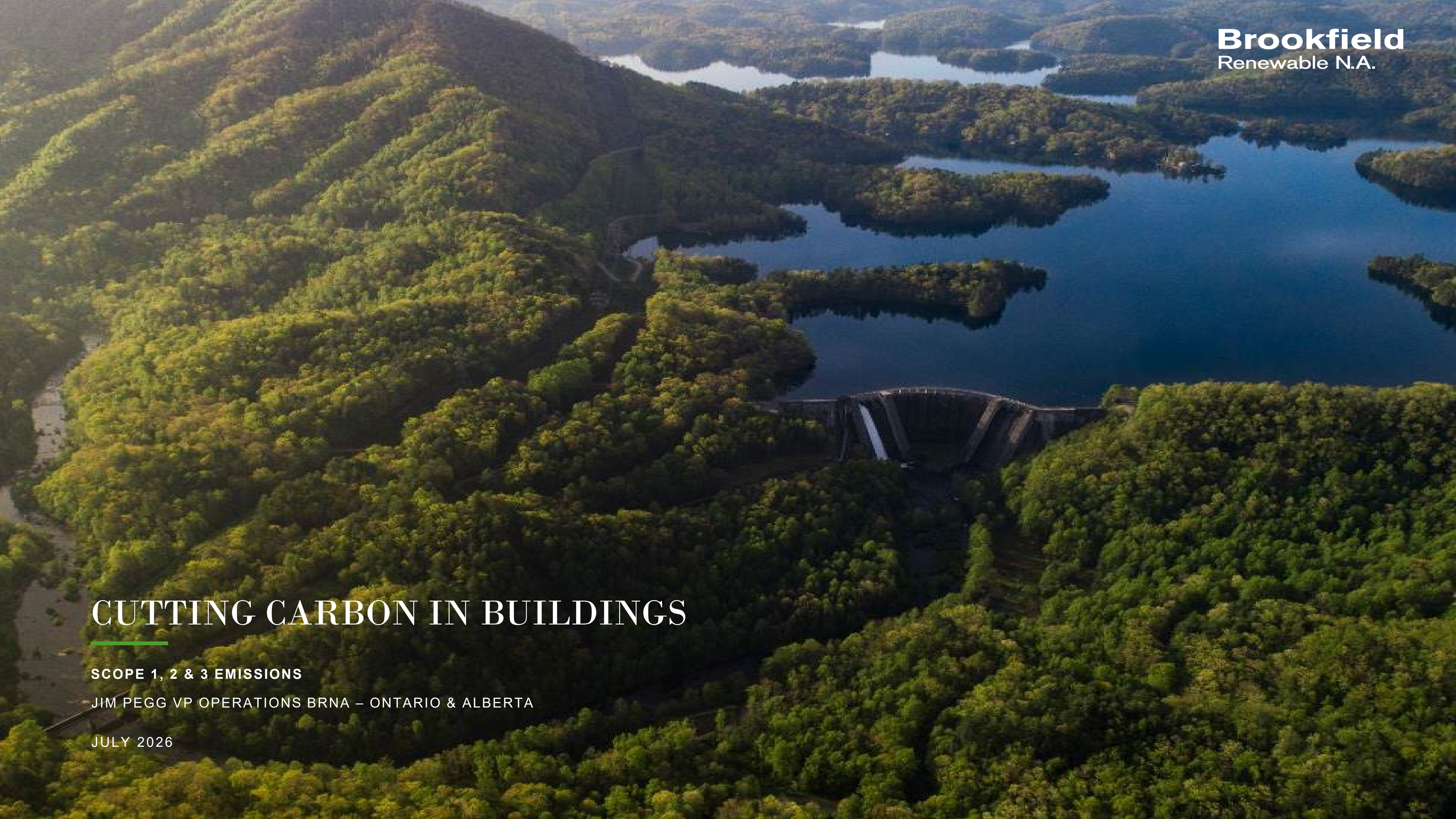

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CUTTING CARBON IN BUILDINGS

SCOPE 1, 2 & 3 EMISSIONS

JIM PEGG VP OPERATIONS BRNA – ONTARIO & ALBERTA

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Scope 1 – DIRECT EMISSIONS

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Fuel you burn on-site

WHAT IT IS

Greenhouse gases released directly from sources the building owns or controls — mostly fuel combusted on the property.

TYPICAL BUILDING SOURCES

- Natural gas furnaces, boilers and water heaters — usually the largest source in Ontario buildings
- On-site combustion: backup diesel generators, gas cooking
- Fugitive refrigerant leaks from chillers, A/C and heat pumps
- Fuel burned by owned/leased fleet vehicles

Note: Global Warming Potential (GWP) is a standardized metric used to compare the heat-trapping power of different greenhouse gases against carbon dioxide GWP acts as a crucial multiplier to calculate **emission factors** for combustible gases (like natural gas or propane), converting their varied atmospheric impacts into a single, unified unit known as **CO_2 equivalents - GWP for CO_2 = 1**

HOW TO CALCULATE

Activity data × emission factor x GWP = tCO_2e

- Natural gas: cubic metres (from bills) × ≈1.921 kg CO_2e/m^3
- Refrigerant: kg leaked × gas GWP (e.g. R-410A ≈ 2,088)
- Diesel fuel burned(L) x 2.68kg CO_2/L
- Leased vehicles fuel burned(L) x 2.31kg CO_2/L

WAYS TO REDUCE

- Electrify heating with cold-climate heat pumps
- Upgrade to high-efficiency condensing boilers + envelope/insulation
- Switch to low-GWP refrigerants and add leak detection
- Tap Save on Energy retrofit incentives (IESO)
- Electric Vehicles
- Variable drive motors

Sources: GHG Protocol · ECCC Emission Factors · Save on Energy / IESO

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Electricity you purchase

WHAT IT IS

Indirect emissions from the generation of energy you buy and consume — purchased electricity, district heating, steam or cooling.

THE ONTARIO ADVANTAGE

– Ontario's grid is typically over 80% non-emitting (hydro, nuclear, wind, solar) — a low electricity emission factor

– Natural-gas plants run mostly at peak hours, so timing of use matters

– Use TAF's Ontario emission factors, updated annually, for accurate accounting

HOW TO CALCULATE

Electricity used (kWh) × grid emission factor = tCO_2e

– Location-based: kWh × Ontario grid factor (varies yearly — 0.039kg CO_2/kWh)

– Market-based: apply contracts / RECs (Renewable Energy Credits) you purchase

WAYS TO REDUCE

– Cut demand: LED lighting, smart controls and building automation

– Generate on-site with rooftop solar PV or wind

– Shift load off peak hours and join demand-response programs or make use of battery storage

– Choose ENERGY STAR equipment; pursue Save on Energy rebates

– Purchase Renewable Energy Credits

Sources: GHG Protocol Scope 2 Guidance · TAF Ontario Emissions Factors · IESO

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Indirect Emissions

WHAT IT IS

All other indirect emissions across the building's
Often the largest share, and the hardest to measure.

COMMON BUILDING SOURCES

- Tenant energy use, employee & visitor commuting
- Water, wastewater and operational waste
- Purchased goods, services and business travel

HOW TO CALCULATE

Screen categories, then estimate the material ones

- Activity-based: quantity × category factor (waste, travel, water)
- Embodied carbon: whole-building life-cycle assessment (CAGBC / NRC guidance)

WAYS TO REDUCE

- Specify low-carbon concrete and reused / recycled materials
- Divert waste through recycling and composting; improve water efficiency
- Use green leases to align tenants on energy data and targets
- Add EV charging and transit incentives; engage suppliers

Sources: GHG Protocol Scope 3 Standard · CAGBC · NRC Whole-Building LCA Guide

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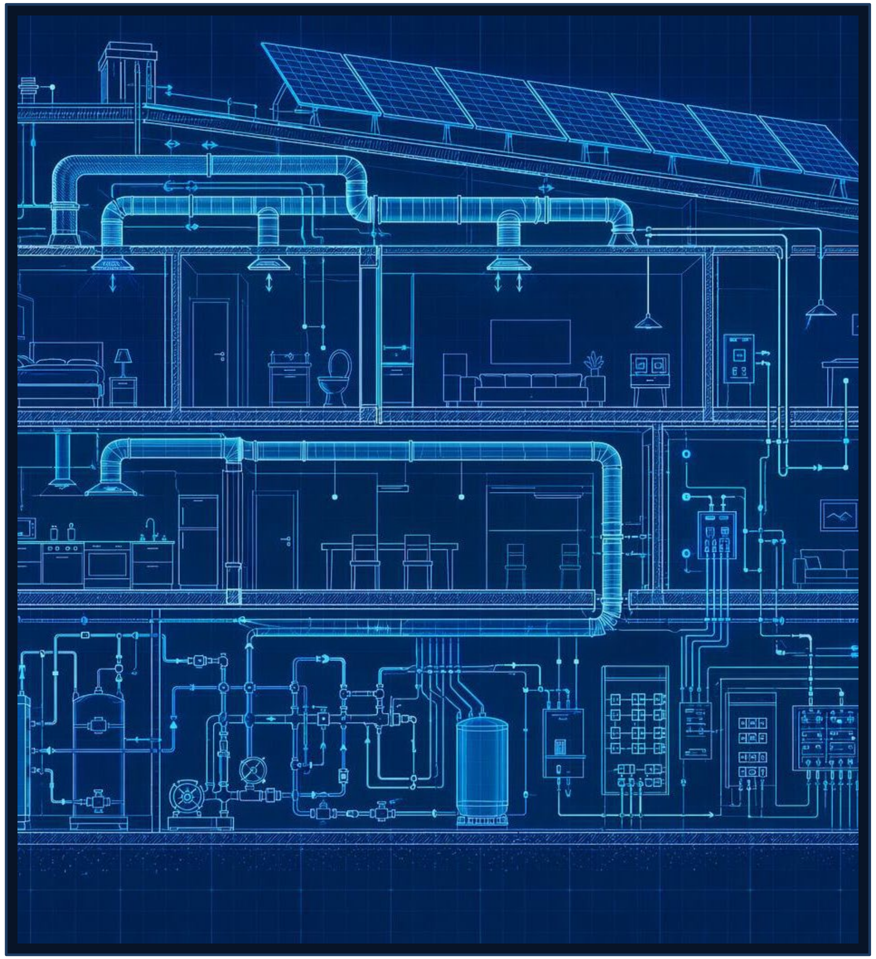
Calculating total building emissions



Total emissions (tCO_2e) = Σ (activity data × emission factor)

…summed across Scope 1 + Scope 2 + Scope 3, with every gas converted to CO_2e using its GWP (CH_4 ≈ 28×, N_2O ≈ 265×).

1. **Set the boundary** Use the operational-control approach (GHG Protocol / ISO 14064).
2. **Gather data (monthly)** Gas (m³), electricity (kWh), refrigerant logs, fleet fuel, water, waste.
3. **Apply emission factors** Environment Climate Change Canada for natural gas, The Atmospheric Fund for Ontario electricity, Inter Governmental Panel on Climate Change GWPs for other gases.
4. **Convert & total by scope** Turn each gas into CO_2e and sum Scope 1, 2 and 3 into one figure.
5. **Lock a base year** Record sources and assumptions so you can track reductions over time.

Ontario: O. Reg. 390/18 requires reporting & verification at ≥10,000 tCO_2e/yr

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From baseline to lower carbon

SCOPE 1
Direct

Fuel burned on-site — gas heat, generators, refrigerant leaks, fleet.

SCOPE 2
Energy

Purchased electricity & heat — low-carbon on Ontario's clean grid.

SCOPE 3
Value chain

Embodied carbon, tenants, commuting, waste, water, suppliers.

TAKEAWAYS

- A clean electricity grid makes heating electrification a powerful lever
- On-site natural gas is usually the biggest emissions source to tackle
- Embodied carbon matters most for new construction and major retrofits
- You can't cut what you don't measure — start with a solid baseline

YOUR ACTION PLAN

- ✔ Measure a baseline year
- ✔ Set targets and prioritize low-cost efficiency wins
- ✔ Electrify heating; add solar and load shifting
- ✔ Tap Save on Energy / IESO incentives to fund it

Frameworks: GHG Protocol · ISO 14064 · ECCC & TAF emission factors

Example Scenario

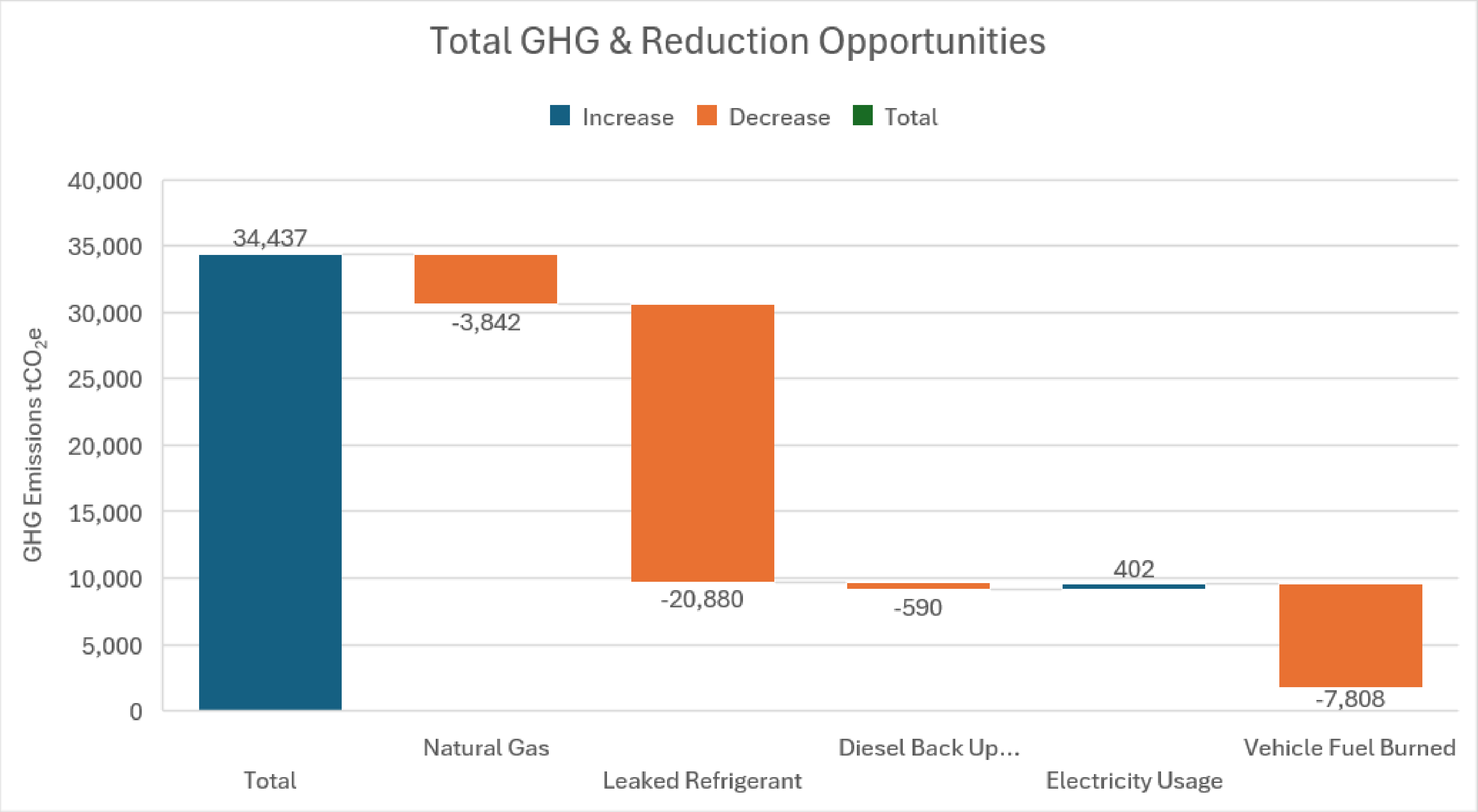
Boundary: Building

Example Data			
Activity	**Activity Data**	**Emission Factor**	**GWP**
Natural Gas	2216 cubic metres per year	1.921 kg CO_2e/m^3	1
Leaked Refrigerant	10kG per year	10kg CH_4	2,088
Diesel Back Up Generator	320L of diesel burned	2.68kg CO_2/L	1
Electricity Usage	16,283 kWh in a year	0.039kg CO_2/kWh	1
Vehicle Fuel Burned	3,380L in a year	2.31kg CO_2/L	1

Example Calculation

Activity	Data	Emission Factor	GWP	Total	
Natural Gas	2216	1.921	1	4,257	
Leaked Refrigerant	10	10	2088	20,880	
Diesel Back Up Generator	320	2.68	1	858	
Electricity Usage	16283	0.039	1	635	
Vehicle Fuel Burned	3380	2.31	1	7,808	
Total GHG				34,437	tCO_2e

Example Scenario



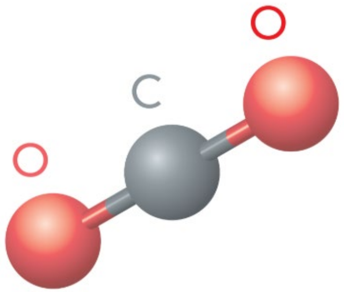
Example GHG Reduction Opportunities

- Convert natural gas devices to electric
- Repair leaking refrigerant
- Reduce diesel back up, use battery back ups
- Switch to electric vehicles

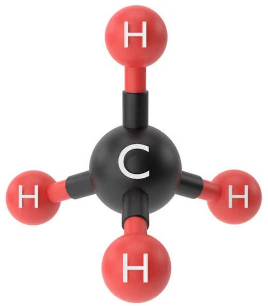
New Total GHG Emissions 1,720 tCO_2e

Examples of GHGs

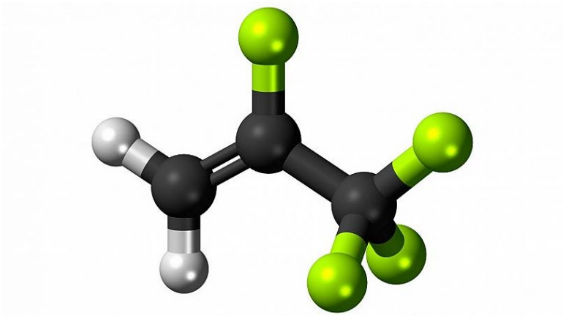
Carbon Dioxide
(CO_2)



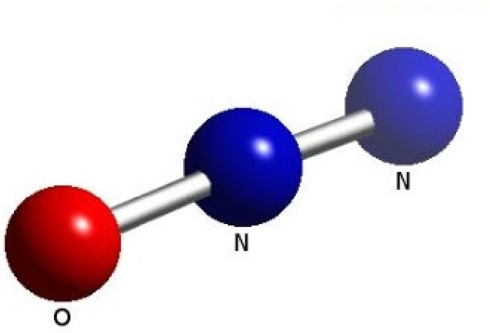
Methane
(CH_4)



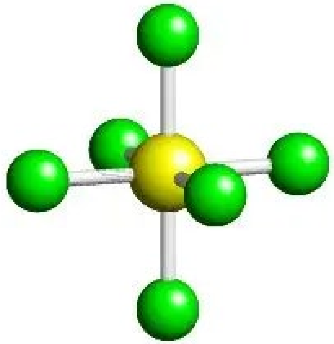
Hydrofluorocarbons
(HFC_S)



Nitrous Oxide
(N_2O)



Sulfur Hexafluoride
(SF_6)



Perfluorocarbons
(PFC_S)